📘 MOLE CONCEPT – COPYABLE NOTES (NO LATEX)

1. Mole kya hota hai

1 mole = 6.022 × 10^23 particles

Isko Avogadro Number bolte hain

2. Basic Formula

n = m / M

n = number of moles

m = mass (grams me)

M = molar mass (g/mol)

3. Number of Particles

N = n × NA

NA = 6.022 × 10^23

Combined formula:

N = (m / M) × NA

4. Gas (STP pe)

n = V / 22.4

V = volume in liters

1 mole gas = 22.4 L (STP)

5. Density Relation

n = (d × V) / M

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d = density

V = volume

6. Molarity

M = moles / volume (in liters)

7. Limiting Reagent

Step 1: Sabka moles nikalo

Step 2: Ratio compare karo (balanced equation se)

Step 3: Jo kam ho → limiting reagent

8. Percentage Composition

% = (mass of element / molar mass) × 100

9. Empirical Formula

Step:

% ko grams maan lo

moles nikalo

smallest se divide karo

ratio fix karo

10. Molecular Formula

Molecular formula = n × empirical formula

n = molecular mass / empirical mass

🔥 *QUICK REVISION*

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$n = m / M$

$N = n \times NA$

Gas: $n = V / 22.4$

Limiting reagent → product decide karta hai

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